

Course Code	Course Title	L	T	P	C
BECE403E	Embedded Systems Design	3	0	2	4
Pre-requisite	BECE204L, BECE204P	Syllabus version			
		1.0			
Course Objectives					
1. To acquaint students with definition, characteristics, challenges and design lifecycle of Embedded Systems by imparting the fundamental knowledge of I/O interfacing, serial communication protocols, wireless technologies, design using UML models 2. To familiarize the concepts and features of Real-time operating systems, task scheduling, and inter-task communication. 3. To impart various programming tools, modeling and simulation packages to program, design, simulate and build Embedded Systems					
Course Outcomes					
At the end of the course, students will be able to 1. Design any application, based on the given specifications by keeping in mind different design metrics. 2. Apply the skills attained to differentiate Microprocessor/Microcontroller and interface various peripherals for a particular application. 3. Demonstrate proficiency in using device drivers, firmware and debugging tools. 4. Analyze the specific perspective of the embedded application using different modelling languages 5. Compare and contrast various wired and wireless protocols 6. Explore the concepts of RTOS and apply the knowledge for developing real-time systems					
Module:1		Embedded System Product Development			4 hours
Characteristics of embedded systems, Classification of embedded systems, Embedded product development cycle, Embedded System Design Challenges, Performance and Benchmarking Tools.					
Module:2		Embedded Hardware Design			5 hours
Processor classification - general purpose, customized, application specific processors, Microcontroller architectures (RISC, CISC), Embedded Memory, Strategic selection of processor and memory, Power Supply Design Considerations for Embedded Systems.					
Module:3		Embedded Software Development Environment			6 hours
Cross assemblers/compilers, Linker, Runtime Library, Pre-processor Workflow, make files, Compiler Tool chains – gcc & ARM, Device Driver, Firmware, Middleware - Debugging tools: Emulators, Simulators, In-Circuit Debuggers, Logic Analyzer, Integrated Development Environment (IDE).					
Module:4		Modeling Embedded Systems			6 hours
Control data flow graph, Finite state machine model, Petrinet Model, Unified model language					
Module:5		Programming the Peripherals of Microcontrollers			6 hours
Programming GPIO pins, Timers / Counters, Watchdog Timer, PWM generation, ADC, DAC, LED, switches, keypad, LCD.					
Module:6		Emerging Communication Protocols			8 hours

UART, SPI, I2C, NFC, CAN, Bluetooth, Zigbee, Wi-Fi			
Module:7		Embedded Real –Time Operating Systems	
		8 hours	
Introduction to basic concepts of RTOS- Task, process & threads, Multiprocessing and Multitasking, Preemptive and non-preemptive scheduling, Schedulability Analysis, Inter process Communication, Performance Metrics of RTOS			
Module:8		Contemporary Issues	
		2 hours	
		Total Lecture hours:	
		45 hours	
Text Book(s)			
1.	Raj Kamal, “Embedded systems Architecture, Programming and Design”, 2017, Third Edition, McGraw Hill Education, India.		
Reference Books			
1.	Marilyn Wolf, “Computers as components: Principles of Embedded Computing System Design”, 2017, Fourth Edition, Morgan Kaufmann publications (Elsevier), United States.		
2.	Jiacun Wang, "Real-Time Embedded Systems", 2017, First Edition, Wiley Publishers, United States.		
Mode of Evaluation: Continuous Assessment Test, Digital Assignment, Quiz and Final Assessment Test			
Indicative Experiments			
1.	Experiments based on interfacing I/O devices		4 hours
2.	Experiments based on monitoring and control using sensors and actuators		6 hours
3.	Experiments based on wired Communications Protocols (UART, SPI, I2C, CAN)		8 hours
4.	Experiments based wireless Communications Protocols (Wi-Fi, Bluetooth)		6 hours
5.	Experiments based on RTOS		6 hours
Total Laboratory Hours			30 hours
Mode of assessment: Continuous assessment and FAT			
Recommended by Board of Studies		28-02-2023	
Approved by Academic Council		No. 69	Date 16-03-2023